



1

Solutions to Initial Centroids Problem

- Multiple runs
 - Helps, but probability is not on your side
- Sample and use hierarchical clustering to determine initial centroids
- Select more than k initial centroids and then select among these initial centroids
 - Select most widely separated
- Postprocessing
- Generate a larger number of clusters and then perform a hierarchical clustering
- Bisecting K-means
 - Not as susceptible to initialization issues

2

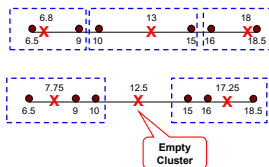
K-means++

- This approach can be slower than random initialization, but very consistently produces better results in terms of SSE
- The k-means++ algorithm guarantees an approximation ratio $O(\log k)$ in expectation, where k is the number of centers
- To select a set of initial centroids, C , perform the following
 1. Select an initial point at random to be the first centroid
 2. For $k - 1$ steps
 3. For each of the N points, x_i , $1 \leq i \leq N$, find the minimum squared distance to the currently selected centroids, $C_1, \dots, C_j$, $1 \leq j < k$, i.e., $\min d^2(C_j, x_i)$
 4. Randomly select a new centroid by choosing a point with probability proportional to $\frac{\min d^2(C_j, x_i)}{\sum \min d^2(C_j, x_i)}$
 5. End For

3

Empty Clusters

- K-means can yield empty clusters



4

Handling Empty Clusters

- Basic K-means algorithm can yield empty clusters
- Several strategies
 - Choose the point that contributes most to SSE
 - Choose a point from the cluster with the highest SSE
 - If there are several empty clusters, the above can be repeated several times.

5

Updating Centers Incrementally

- In the basic K-means algorithm, centroids are updated after all points are assigned to a centroid
- An alternative is to update the centroids after each assignment (incremental approach)
 - Each assignment updates zero or two centroids
 - More expensive
 - Introduces an order dependency
 - Never get an empty cluster
 - Can use "weights" to change the impact

6

Pre-processing and Post-processing

- Pre-processing
 - Normalize the data
 - Eliminate outliers
- Post-processing
 - Eliminate small clusters that may represent outliers
 - Split 'loose' clusters, i.e., clusters with relatively high SSE
 - Merge clusters that are 'close' and that have relatively low SSE
 - Can use these steps during the clustering process
 - ISODATA

7

Bisecting K-means

- Bisecting K-means algorithm
 - Variant of K-means that can produce a partitional or a hierarchical clustering

```

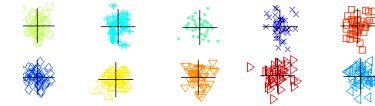
1: Initialize the list of clusters to contain the cluster containing all points.
2: repeat
3:   Select a cluster from the list of clusters
4:   for  $i = 1$  to  $number\_of\_iterations$  do
5:     Bisect the selected cluster using basic K-means
6:   end for
7:   Add the two clusters from the bisection with the lowest SSE to the list of clusters.
8: until Until the list of clusters contains  $K$  clusters

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CLUTO: <http://glaros.dtc.umn.edu/uhtml/home/cluto/clutooverview>

8

Bisecting K-means Example



9

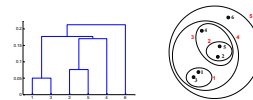


HIERARCHICAL CLUSTERING

10

Hierarchical Clustering

- Produces a set of nested clusters organized as a hierarchical tree
- Can be visualized as a dendrogram
 - A tree like diagram that records the sequences of merges or splits



11

Strengths of Hierarchical Clustering

- Do not have to assume any particular number of clusters
 - Any desired number of clusters can be obtained by 'cutting' the dendrogram at the proper level
- They may correspond to meaningful taxonomies
 - Example in biological sciences (e.g., animal kingdom, phylogeny reconstruction, ...)

12

Hierarchical Clustering

- Two main types of hierarchical clustering
 - **Agglomerative:**
 - Start with the points as individual clusters
 - At each step, merge the closest pair of clusters until only one cluster (or k clusters) left
 - **Divisive:**
 - Start with one, all-inclusive cluster
 - At each step, split a cluster until each cluster contains an individual point (or there are k clusters)
- Traditional hierarchical algorithms use a similarity or distance matrix
- Merge or split one cluster at a time

13

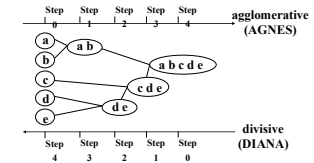
Hierarchical Clustering

- Clusters are created in levels actually creating sets of clusters at each level.
- **Agglomerative**
 - Initially each item in its own cluster
 - Iteratively clusters are merged together
 - Bottom Up
- **Divisive**
 - Initially all items in one cluster
 - Large clusters are successively divided
 - Top Down

14

Hierarchical Clustering

- Use distance matrix as clustering criteria. This method does not require the number of clusters k as an input, but needs a termination condition



15

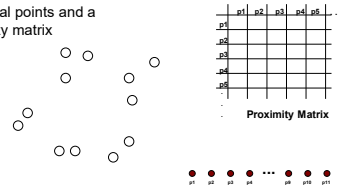
Agglomerative Clustering Algorithm

- Most popular hierarchical clustering technique
- Basic algorithm is straightforward
 1. Compute the proximity matrix
 2. Let each data point be a cluster
 3. **Repeat**
 4. Merge the two closest clusters
 5. Update the proximity matrix
 6. **Until** only a single cluster remains
- Key operation is the computation of the proximity of two clusters
 - Different approaches to defining the distance between clusters distinguish the different algorithms

16

Starting Situation

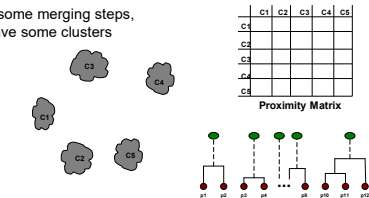
- Start with clusters of individual points and a proximity matrix



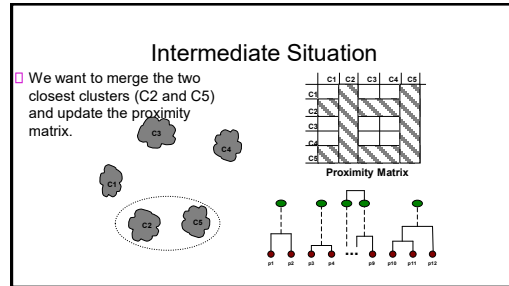
17

Intermediate Situation

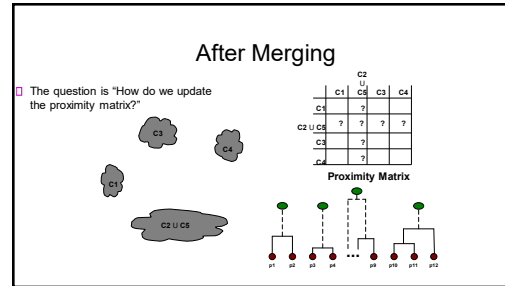
- After some merging steps, we have some clusters



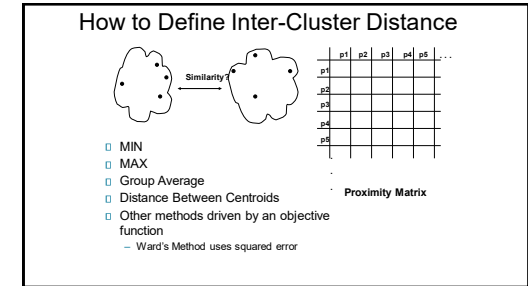
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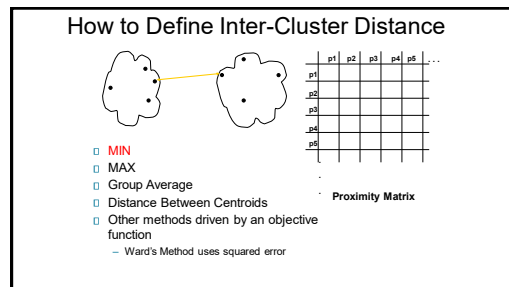
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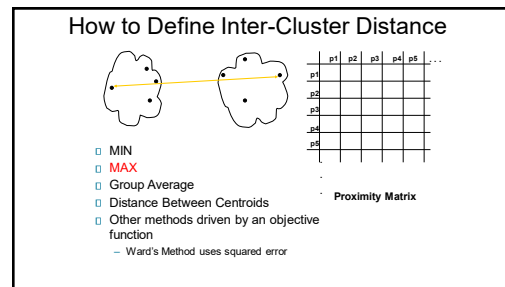
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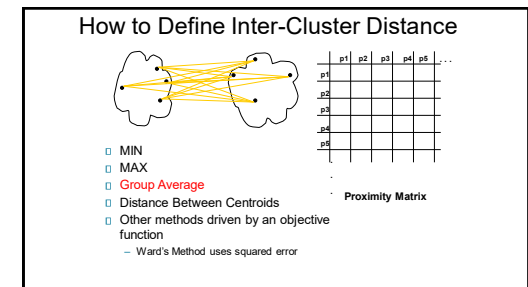
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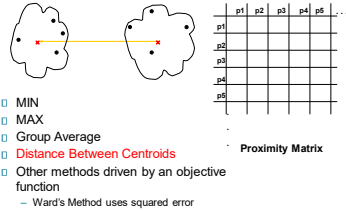


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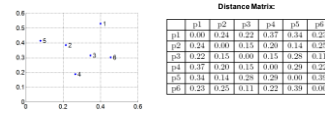
How to Define Inter-Cluster Distance



25

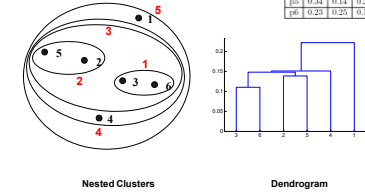
MIN or Single Link

- Proximity of two clusters is based on the two closest points in the different clusters
- Determined by one pair of points, i.e., by one link in the proximity graph
- Example:



26

Hierarchical Clustering: MIN

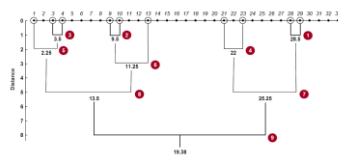


27

Single Link - Example

Agglomerative clustering example

- Perform agglomerative clustering on the following dataset using the single-linkage metric
 - $X = \{1, 3, 4, 9, 10, 13, 21, 23, 28, 29\}$
 - In case of ties, always merge the pair of clusters with the largest mean
 - Indicate the order in which the merging operations occur

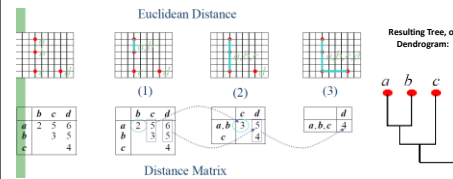


28

Clustering Algorithms

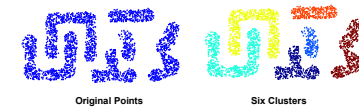
Hierarchical Clustering:

Example: Single-Link (Minimum) Method:



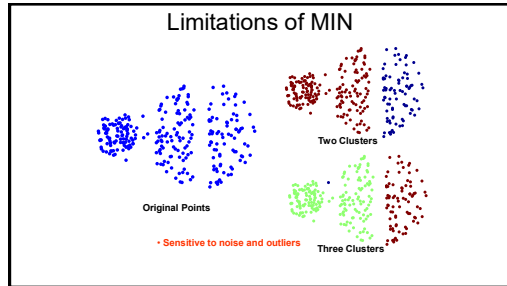
29

Strength of MIN

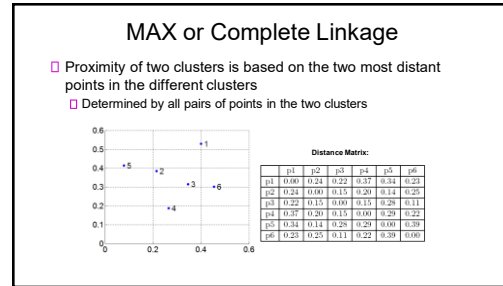


Can handle non-elliptical shapes

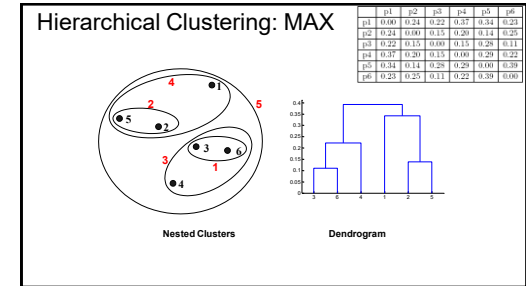
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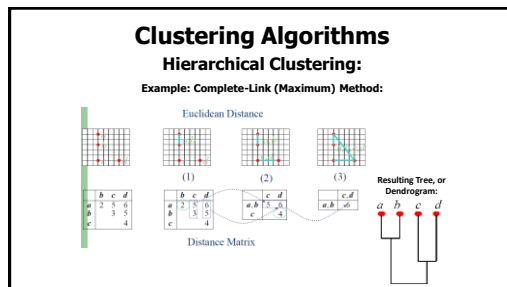
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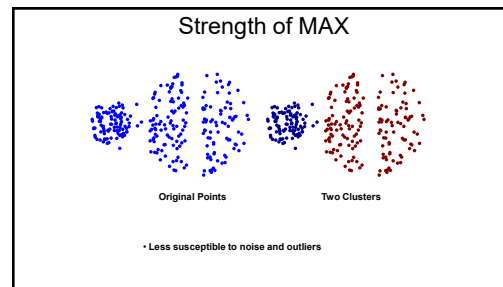
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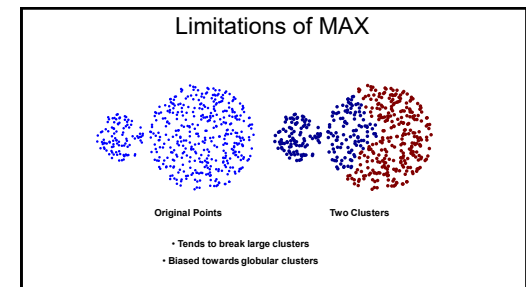
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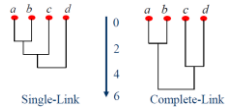


36

Hierarchical Clustering:

In a dendrogram, the *length of each tree branch* represents the distance between clusters it joins.

Different dendrograms may arise when different Linkage methods are used.



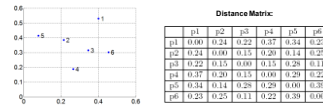
37

Group Average

- Proximity of two clusters is the average of pairwise proximity between points in the two clusters.

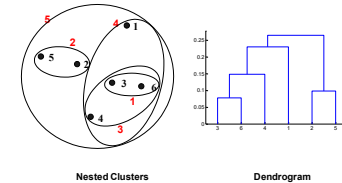
$$\text{proximity}(\text{Cluster}_i, \text{Cluster}_j) = \frac{\sum_{p_i \in \text{Cluster}_i, p_j \in \text{Cluster}_j} \text{proximity}(p_i, p_j)}{|\text{Cluster}_i| \cdot |\text{Cluster}_j|}$$

- Need to use average connectivity for scalability since total proximity favors large clusters



38

Hierarchical Clustering: Group Average



39

Hierarchical Clustering: Group Average

- Compromise between Single and Complete Link
- Strengths
 - Less susceptible to noise and outliers
- Limitations
 - Biased towards globular clusters

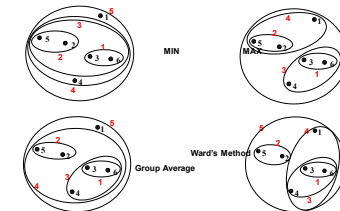
40

Cluster Similarity: Ward's Method

- Similarity of two clusters is based on the increase in squared error when two clusters are merged
 - Similar to group average if distance between points is distance squared
- Less susceptible to noise and outliers
- Biased towards globular clusters
- Hierarchical analogue of K-means
 - Can be used to initialize K-means

41

Hierarchical Clustering: Comparison

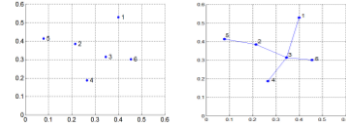


42

MST: Divisive Hierarchical Clustering

□ Build MST (Minimum Spanning Tree)

- Start with a tree that consists of any point
- In successive steps, look for the closest pair of points (p, q) such that one point (p) is in the current tree but the other (q) is not
- Add q to the tree and put an edge between p and q



43

MST: Divisive Hierarchical Clustering

- Use MST for constructing hierarchy of clusters

Algorithm 7.5 MST Divisive Hierarchical Clustering Algorithm

- 1: Compute a minimum spanning tree for the proximity graph.
- 2: **repeat**
- 3: Create a new cluster by breaking the link corresponding to the largest distance (smallest similarity).
- 4: **until** Only singleton clusters remain

44

Hierarchical Clustering: Time and Space requirements

- $O(N^2)$ space since it uses the proximity matrix.
 - N is the number of points.
- $O(N^3)$ time in many cases
 - There are N steps and at each step the size, N^2 , proximity matrix must be updated and searched
 - Complexity can be reduced to $O(N^2 \log(N))$ time with some cleverness

45

Hierarchical Clustering: Problems and Limitations

- Once a decision is made to combine two clusters, it cannot be undone
- No global objective function is directly minimized
- Different schemes have problems with one or more of the following:
 - Sensitivity to noise and outliers
 - Difficulty handling clusters of different sizes and non-globular shapes
 - Breaking large clusters

46

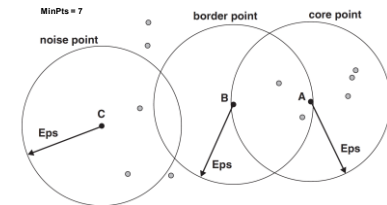
DBSCAN

- DBSCAN is a density-based algorithm.

- Density = number of points within a specified radius (Eps)
- A point is a **core point** if it has at least a specified number of points (MinPts) within Eps
 - These are points that are at the interior of a cluster
 - Counts the point itself
- A **border point** is not a core point, but is in the neighborhood of a core point
- A **noise point** is any point that is not a core point or a border point

47

DBSCAN: Core, Border, and Noise Points



48

DBSCAN Algorithm

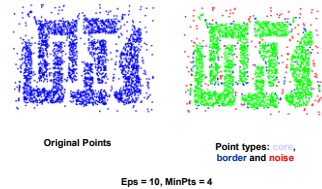
- Eliminate noise points
 - Perform clustering on the remaining points
- ```

current_cluster_label ← 1
for all core points do
 if the core point has no cluster label then
 current_cluster_label ← current_cluster_label + 1
 Label the current core point with cluster label current_cluster_label
 end if
 for all points in the Eps-neighborhood, except i^{th} the point itself do
 if the point does not have a cluster label then
 Label the point with cluster label current_cluster_label
 end if
 end for
end for

```

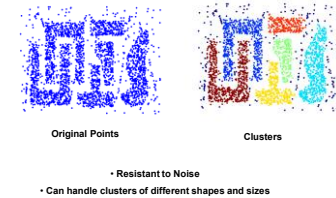
49

## DBSCAN: Core, Border and Noise Points



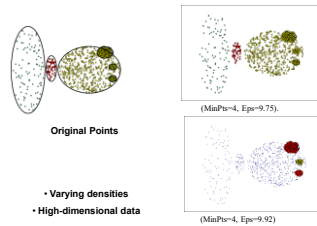
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## When DBSCAN Works Well



51

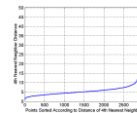
## When DBSCAN Does NOT Work Well



52

## DBSCAN: Determining EPS and MinPts

- Idea is that for points in a cluster, their  $k^{th}$  nearest neighbors are at roughly the same distance
- Noise points have the  $k^{th}$  nearest neighbor at farther distance
- So, plot sorted distance of every point to its  $k^{th}$  nearest neighbor

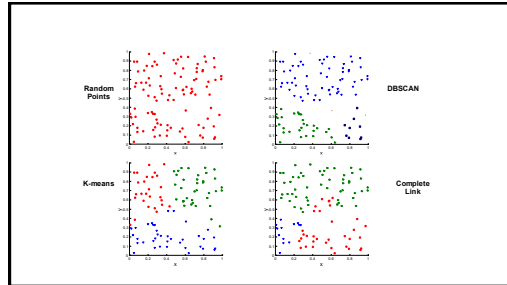


53

## Cluster Validity

- For supervised classification we have a variety of measures to evaluate how good our model is
  - Accuracy, precision, recall
- For cluster analysis, the analogous question is how to evaluate the "goodness" of the resulting clusters?
- But "clusters are in the eye of the beholder"
- Then why do we want to evaluate them?
  - To avoid finding patterns in noise
  - To compare clustering algorithms
  - To compare two sets of clusters
  - To compare two clusters

54



55

### Different Aspects of Cluster Validation

1. Determining the **clustering tendency** of a set of data, i.e., distinguishing whether non-random structure actually exists in the data.
2. Comparing the results of a cluster analysis to externally known results, e.g., to externally given class labels.
3. Evaluating how well the results of a cluster analysis fit the data *without* reference to external information.
  - Use only the data
4. Comparing the results of two different sets of cluster analyses to determine which is better.
5. Determining the 'correct' number of clusters.

For 2, 3, and 4, we can further distinguish whether we want to evaluate the entire clustering or just individual clusters.

56

### Measures of Cluster Validity

- Numerical measures that are applied to judge various aspects of cluster validity, are classified into the following three types.
  - **External Index:** Used to measure the extent to which cluster labels match externally supplied class labels.
    - Entropy
  - **Internal Index:** Used to measure the goodness of a clustering structure *without* respect to external information.
    - Sum of Squared Error (SSE)
  - **Relative Index:** Used to compare two different clusterings or clusters.
    - Often an external or internal index is used for this function, e.g., SSE or entropy
- Sometimes these are referred to as **criteria** instead of **indices**
- However, sometimes criterion is the general strategy and index is the numerical measure that implements the criterion.

57

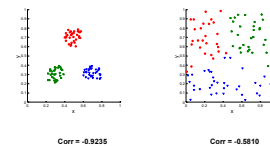
### Measuring Cluster Validity Via Correlation

- Two matrices
  - Proximity Matrix
  - Ideal Similarity Matrix
    - One row and one column for each data point
    - An entry is 1 if the associated pair of points belong to the same cluster
    - An entry is 0 if the associated pair of points belongs to different clusters
- Compute the correlation between the two matrices
  - Since the matrices are symmetric, only the correlation between  $n(n-1)/2$  entries needs to be calculated.
- High correlation indicates that points that belong to the same cluster are close to each other.
- Not a good measure for some density or contiguity based clusters.

58

### Measuring Cluster Validity Via Correlation

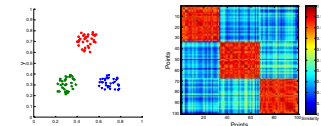
- Correlation of ideal similarity and proximity matrices for the K-means clusterings of the following two data sets.



59

### Using Similarity Matrix for Cluster Validation

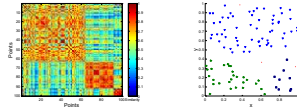
- Order the similarity matrix with respect to cluster labels and inspect visually.



60

## Using Similarity Matrix for Cluster Validation

- Clusters in random data are not so crisp

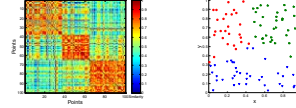


DBSCAN

61

## Using Similarity Matrix for Cluster Validation

- Clusters in random data are not so crisp

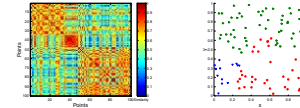


K-means

62

## Using Similarity Matrix for Cluster Validation

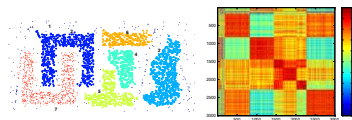
- Clusters in random data are not so crisp



Complete Link

63

## Using Similarity Matrix for Cluster Validation

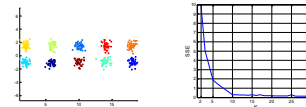


DBSCAN

64

## Internal Measures: SSE

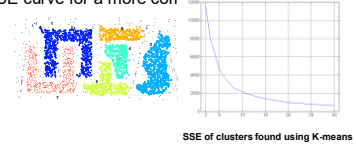
- Clusters in more complicated figures aren't well separated
- Internal Index: Used to measure the goodness of a clustering structure without respect to external information
  - SSE
- SSE is good for comparing two clusterings or two clusters (average SSE).
- Can also be used to estimate the number of clusters



65

## Internal Measures: SSE

- SSE curve for a more complex



SSE of clusters found using K-means

66

### Framework for Cluster Validity

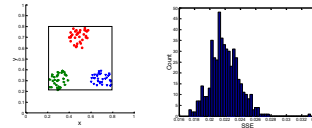
- Need a framework to interpret any measure.
  - For example, if our measure of evaluation has the value, 10, is that good, fair, or poor?
- Statistics provide a framework for cluster validity
  - The more "atypical" a clustering result is, the more likely it represents valid structure in the data
  - Can compare the values of an index that result from random data or clusterings to those of a clustering result.
    - If the value of the index is unlikely, then the cluster results are valid
  - These approaches are more complicated and harder to understand.
- For comparing the results of two different sets of cluster analyses, a framework is less necessary.
  - However, there is the question of whether the difference between two index values is significant

67

### Statistical Framework for SSE

#### □ Example

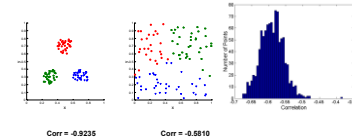
- Compare SSE of 0.005 against three clusters in random data
- Histogram shows SSE of three clusters in 500 sets of random data points of size 100 distributed over the range 0.2 – 0.8 for x and y values



68

### Statistical Framework for Correlation

- Correlation of ideal similarity and proximity matrices for the K-means clusterings of the following two data sets.



69

### Internal Measures: Cohesion and Separation

- **Cluster Cohesion:** Measures how closely related are objects in a cluster
  - Example: SSE
- **Cluster Separation:** Measure how distinct or well-separated a cluster is from other clusters
- Example: Squared Error
  - Cohesion is measured by the within cluster sum of squares (SSE)
 
$$SSE = WSS = \sum_i \sum_{x \in C_i} (x - m_i)^2$$
  - Separation is measured by the between cluster sum of squares
 
$$BSS = \sum_i |C_i| (m_i - \bar{m})^2$$

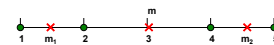
~ Where  $|C_i|$  is the size of cluster  $i$

70

### Internal Measures: Cohesion and Separation

#### □ Example: SSE

- $BSS + WSS = \text{constant}$



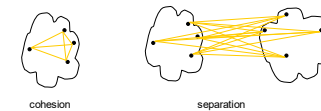
**K=1 cluster:**  $SSE = WSS = (1-3)^2 + (2-3)^2 + (4-3)^2 + (5-3)^2 = 10$   
 $BSS = 4 \times (3-3)^2 = 0$   
 Total = 10 + 0 = 10

**K=2 clusters:**  $SSE = WSS = (1-1.5)^2 + (2-1.5)^2 + (4-4.5)^2 + (5-4.5)^2 = 1$   
 $BSS = 2 \times (3-1.5)^2 + 2 \times (4.5-3)^2 = 9$   
 Total = 1 + 9 = 10

71

### Internal Measures: Cohesion and Separation

- A proximity graph based approach can also be used for cohesion and separation.
  - Cluster cohesion is the sum of the weight of all links within a cluster.
  - Cluster separation is the sum of the weights between nodes in the cluster and nodes outside the cluster.



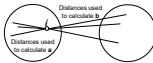
72

### Internal Measures: Silhouette Coefficient

- Silhouette coefficient combines ideas of both cohesion and separation, but for individual points, as well as clusters and clusterings
- For an individual point,  $i$ 
  - Calculate  $a$  = average distance of  $i$  to the points in its cluster
  - Calculate  $b$  = min (average distance of  $i$  to points in another cluster)
  - The silhouette coefficient for a point is then given by

$$s = (b - a) / \max(a, b)$$

- Typically between 0 and 1.
- The closer to 1 the better.



- Can calculate the average silhouette coefficient for a cluster or a clustering

73

### External Measures of Cluster Validity: Entropy and Purity

Table 5.9. K-means Clustering Results for LA Document Data Set

| Cluster | Entertainment | Financial | Foreign | Metro | National | Sports | Entropy | Purity |
|---------|---------------|-----------|---------|-------|----------|--------|---------|--------|
| 1       | 3             | 5         | 40      | 306   | 96       | 27     | 1.2270  | 0.7474 |
| 2       | 4             | 7         | 240     | 29    | 39       | 2      | 1.1472  | 0.7734 |
| 3       | 1             | 1         | 1       | 7     | 4        | 671    | 0.1513  | 0.9796 |
| 4       | 10            | 162       | 3       | 119   | 73       | 2      | 1.7487  | 0.4390 |
| 5       | 331           | 22        | 5       | 70    | 13       | 23     | 1.3976  | 0.7134 |
| 6       | 3             | 358       | 12      | 212   | 48       | 13     | 1.5523  | 0.5323 |
| Total   | 354           | 555       | 341     | 943   | 273      | 728    | 1.1450  | 0.7203 |

**entropy** For each cluster, the class distribution of the data is calculated first, i.e., for cluster  $j$  we compute  $p_{ij}$ , the 'probability' that a member of cluster  $j$  belongs to class  $i$  as follows:  $p_{ij} = m_{ij}/m_j$ , where  $m_j$  is the number of values in cluster  $j$  and  $m_{ij}$  is the number of values of class  $i$  in cluster  $j$ . Then using this class distribution, the entropy of each cluster  $j$  is calculated using the standard formula  $e_j = -\sum_{i=1}^L p_{ij} \log p_{ij}$ , where the  $L$  is the number of classes. The total entropy for a set of clusters is calculated as the sum of the entropies of each cluster weighted by the size of each cluster, i.e.  $e = \sum_{j=1}^K m_j e_j$ , where  $m_j$  is the size of cluster  $j$ ,  $K$  is the number of clusters, and  $m$  is the total number of data points.

**purity** Using the terminology derived for entropy, the purity of cluster  $j$  is given by  $\text{purity}_j = \max_i p_{ij}$  and the overall purity of a clustering is  $\text{purity} = \sum_{j=1}^K \frac{m_j}{m} \text{purity}_j$ .

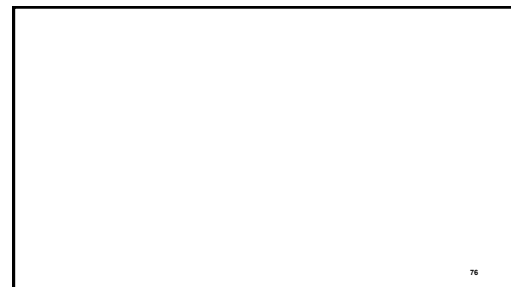
74

Final Comment on Cluster Validity  
"The validation of clustering structures is the most difficult and frustrating part of cluster analysis."

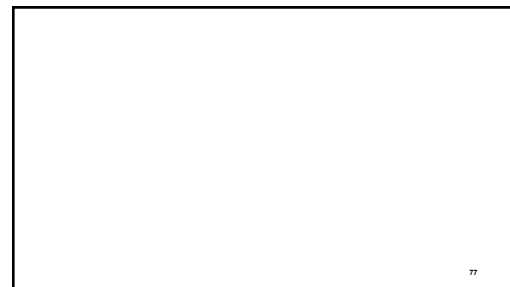
Without a strong effort in this direction, cluster analysis will remain a black art accessible only to those true believers who have experience and great courage."

*Algorithms for Clustering Data*, Jain and Dubes

75



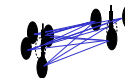
76



77

### Distance Between Clusters

- Single Link**: smallest distance between points. The minimum of all pairwise distances between points in the two clusters
  - Tends to produce long, "loose" clusters
- Complete Link**: largest distance between points. The maximum of all pairwise distances between points in the two clusters
  - Tends to produce very tight clusters
- Average Link**: the average of all pairwise distances between points in the two clusters
- Centroid Linage**: distance between centroids



78

### Hierarchical Algorithms

- Single Link
- Complete Link
- Average Link
- MST Single Link